Queries and Retrieval

Corpus
no queries, no labels

Stage A: Query Generation

Write queries a user of the corpus could ask

Prompt LLM with instruction t (optional) and 3 sampled documents → *What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?*

$q_j = \text{Write}(t, d_1, d_2, d_3)$, 3–4,000 chars, 100 per task

Stage B: Retrieval Pool

Pool every model's top 10 for each query

Qwen3-Embedding-8B	✓	✓	✓	✓	✓	✓
Linq-Embed-Mistral	✓	✓	✓	✓	✗	✓
all-MiniLM-L6-v2	✓	✗	✗	✗	✗	✗
baseline-bm25s	✓	✓	✓	✗	✓	✗

4 of 28 models; ✓ = in the model's top 10 for q_j

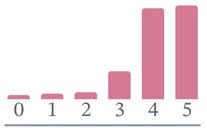
$D_j = \bigcup_m \text{top10}_m(q_j)$, $|D_j| = 87$

each pooled document is judged once

Decision-Model Judge

Read the judge's probability for each answer, instead of sampling one answer

Pointwise (default): grade each (q_j, d) in D_j

(q_j, d) → **judge** → 

$p(k \mid q_j, d)$ over grades 0 to 5

grade $g(q_j, d) = \sum_k k \cdot p(k \mid q_j, d) = 4.3$

grades of the top 10: A  B 

Pairwise: is A's top 10 or B's better?

order shown	P(A better) + ½P(tie)
A first, then B	0.97
B first, then A	0.62

s_{AB} = mean over the two orders = 0.80

(another query) the swap cancels position bias

Gym Score

Turn the judgments into one score per model, comparable across queries and tasks

Pointwise: nDCG@10 over the pool

$$\text{nDCG@10}(m, q_j) = \frac{\sum_{r \le 10} g(q_j, d_{m,r}) \,/\, \log_2(r+1)}{\text{the same for the 10 best in } D_j}$$

here: A = 13.2 / 16.4 = 0.81, B = 5.8 / 16.4 = 0.35

$\text{score}_m = 100 \times$ mean over queries, then over tasks

Pairwise: Bradley–Terry over all s

$P(m \succ n) = \dfrac{\pi_m}{\pi_m + \pi_n}$, $\text{Elo}_m = 400 \log_{10} \pi_m$, mean 1000

Resulting gym ranking (28 models)

1	Qwen3-Embedding-8B	82.7
2	Qwen3-Embedding-4B	81.6
3	Linq-Embed-Mistral	81.2
4	harrier-oss-v1-0.6b	80.2
⋮		
25	all-MiniLM-L6-v2	65.9

Spearman ρ = 0.95 with MTEB's ranking